Prasham Shah

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Education

Harvard College **May 2029**
A.B. Computer Science & Mathematics *Cambridge, MA*
Coursework (ongoing): Math 55a/b (Hon. Linear Algebra, Group Theory, Real Analysis & Complex Analysis), Stat 110 (Probability), Stat 111 (Statistical Inference), CS 51 (Abstraction in Computation), CS 61 (Systems Programming and Machine Organization)

Experience

Dorm to Door Student Storage | *React, Firebase, Stripe, Node.js, TypeScript* **2025 – 2026**
Software Engineer *Remote*

- Architected and shipped full-stack platform for summer-storage startup, scaled from operating across 3 universities to now **nearing 50 campuses nationwide**, owning both the React frontend and Node.js backend, integrating Firebase, Stripe & OAuth
- Engineered delivery-routing infrastructure using **graph-based optimization algorithms** (Clarke-Wright heuristic, 2-opt local optimization, Map APIs, etc.) to minimize route cost and coordinate complex, multi-stop pickup/drop-off schedules at scale
- Built out admin systems, real-time database sync, and authentication with an emphasis on accessibility and mobile UX, supporting **live production traffic** across the full 2025 season (**99.97% uptime**)

Research

Symbolic Regression — Drew University | *Scikit-learn, NumPy, SymPy, Pandas, Python* **Summer 2025**
Lead Machine Learning Researcher *Madison, NJ*

- Led CS research team as **1st author** on IEEE paper, winning **Best Poster at MIT URTC (1st of ∼700 submissions)**
- Trained and evaluated deep **neural networks** to recover closed-form, human-interpretable equations from raw, multivariate physics datasets, exploiting dynamically discovered mathematical symmetries to **reduce dimensionality**
- Designed a novel **adaptive-alpha** technique for the neural-network symmetry-detection subroutine, doubling true-symmetry detection rates and reducing false positives by **10**× over AI Feynman, the prior state-of-the-art system

Noise Resistance — Cambridge Open Engage | *Multimodal LLMs, OpenCV, Statistics, Python, Linux* **2024 – 2025**
Independent Machine Learning Researcher *Remote*

- Benchmarked 3 models (2 MLLMs and 1 LLM), including a **self-hosted LLaVA model**, across **120 custom prompts** (with control, noisy, and classically denoised variants)
- Quantified model robustness to real-world noise types: textual (misspellings) & visual (**Gaussian, salt-and-pepper, speckle**)
- Demonstrated ($\boldsymbol{p < 0.01}$) that MLLMs are significantly less noise-resistant than text-only LLMs on text prompts, **classical denoising pipelines degrade accuracy** on both modalities, and smaller models show disproportionately higher noise sensitivity

Projects

Legal Lieutenant | *Firebase, GPT Fine-tuning, NLP Pipelines, EJS, Express, JavaScript* **2026**

- Led team to develop AI-powered **legal assistant** targeting U.S. immigrants — **1st place** at Empower Hacks 2.0 (**2,000** international competitors, **$210K** prize pool)
- Built **custom NLP chunking pipeline** for legal document ingestion, **fine-tuned GPT-4o-mini** for domain-specific Q&A and summarization, and hand-rolled standards-compliant authentication

SocialScanner | *PyTorch, Computer Vision, NLP, Web Scraping, Parallel Computing, Python* **2025**

- Designed privacy-first, ML-powered **mental-health screening** system for schools — **backed by Northern Burlington HS** administration and presented at a national TSA conference
- Developed **one-model-per-metric architecture** to enable granular activation of only desired signals (social media **text/images**, grades, counselor **audio transcripts**, etc.); used **PyTorch** to build models and **parallel computing** to generate reports in bulk

Ultra Ball | *WebSockets, SvelteKit, Behavior-tree AI, TypeScript* **2024**

- Crafted **real-time multiplayer web game** from scratch — grew organically to **500+ daily active users** months post-launch
- Deployed fully custom, client-agnostic **WebSocket server**, resolving complex interactions between 15+ move types, managing **bot AIs** with randomized personalities and behavior trees, and automating lobby lifecycle

Additional Honors

Lockheed Martin Code Quest: 3rd place, Advanced Division, inter-state algorithmic coding competition
U.S. Presidential Scholar: 1 of 600 national semifinalists (top 0.01% of eligible students)
Eagle Scout: Including Senior Patrol Leader, managing 40+ scouts and multi-month initiatives

Technical Skills

Languages: Python, TypeScript, C++, Java, Rust, Go, SQL, HTML/CSS, Bash
Full-Stack: React / Next.js, SvelteKit, Node / Express, REST APIs, OAuth, WebSockets
AI/ML: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, SymPy, Behavior-tree AI, LLM Fine-tuning
Infrastructure: Firebase / GCP, AWS (S3 & EC2), Git, Linux
Techniques: Parallel Computing, Graph Algorithms, Statistical Benchmarking, Data-driven Optimization